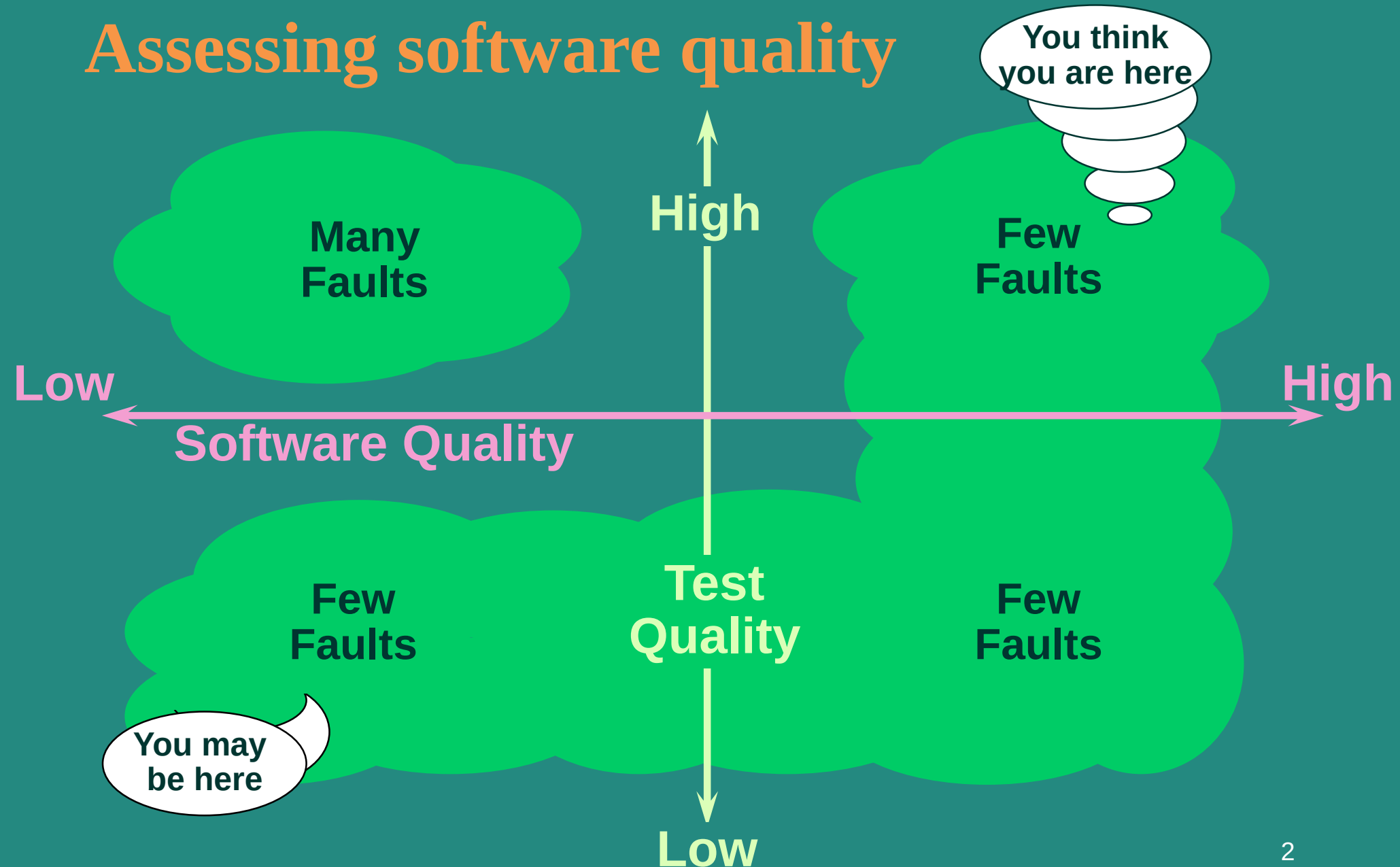


Software Testing

Principles of Testing

***“Testing can only show the presence of errors,
not their absence”***

Assessing software quality



A traditional testing approach

- Show that the system:
 - does what it should
 - doesn't do what it shouldn't

Goal: show working

Success: system works

Fastest achievement: easy test cases



Result: faults left in

A better testing approach

- Show that the system:
 - does what it shouldn't
 - doesn't do what it should

Goal: find faults

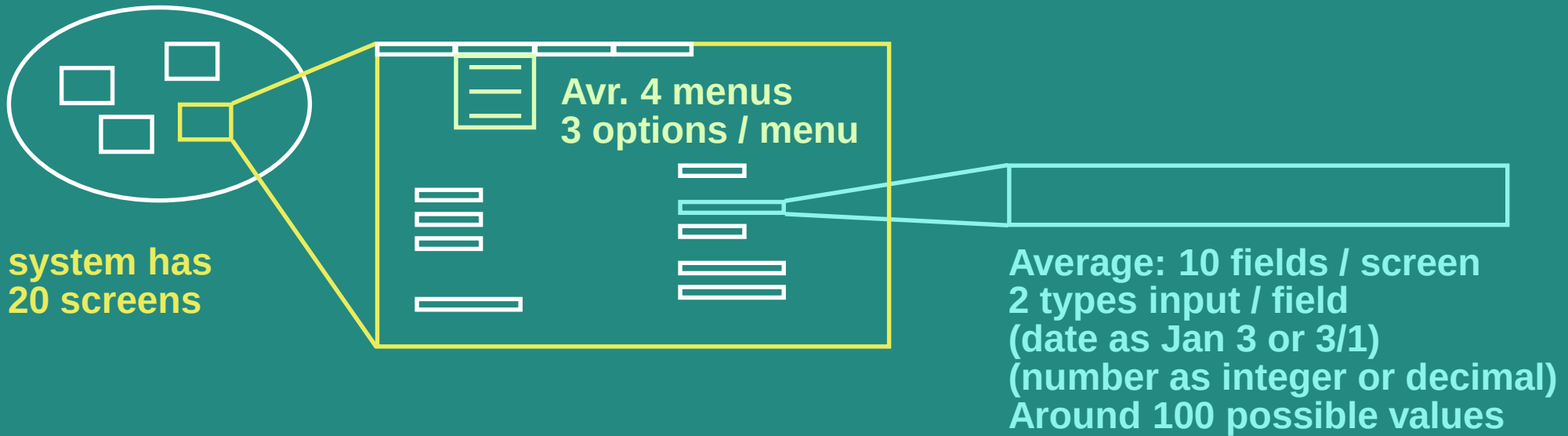
Success: system fails

Fastest achievement: difficult test cases



Result: fewer faults left in

Why not just "test everything"?



Total for 'exhaustive' testing:

$$20 \times 4 \times 3 \times 10 \times 2 \times 100 = 480,000 \text{ tests}$$

If 1 second per test, 8000 mins, 133 hrs, 17.7 days
(not counting finger trouble, faults or retest)

10 secs = 34 wks, 1 min = 4 yrs, 10 min = 40 yrs

How much testing is enough?

- it's never enough ✗
- when you have done what you planned ✗
- when your customer/user is happy ✗
- when you have proved that the system works correctly ✗
- when you are **confident** that the system works correctly ✓
- it depends on the **risks** for your system ✓

Most important principle

Prioritise tests

so that,

**whenever you stop testing,
you have done the best testing
in the time available.**

Seven Testing Principles

1. Testing shows the presence of defects
2. Exhaustive testing is impossible
3. Early testing
4. Defect clustering
5. Pesticide paradox
6. Testing is context dependent
7. Absence-of-errors fallacy

7 Principles of Software Testing

Principle 1	Testing shows presence of defects
Principle 2	Exhaustive testing is impossible
Principle 3	Early Testing
Principle 4	Defect Clustering
Principle 5	Pesticide Paradox
Principle 6	Testing is context dependent
Principle 7	Absence of errors - fallacy

Chỉ ra sự hiện diện của lỗi

Không thể kiểm thử hết mọi trường hợp

Kiểm thử càng sớm càng tốt

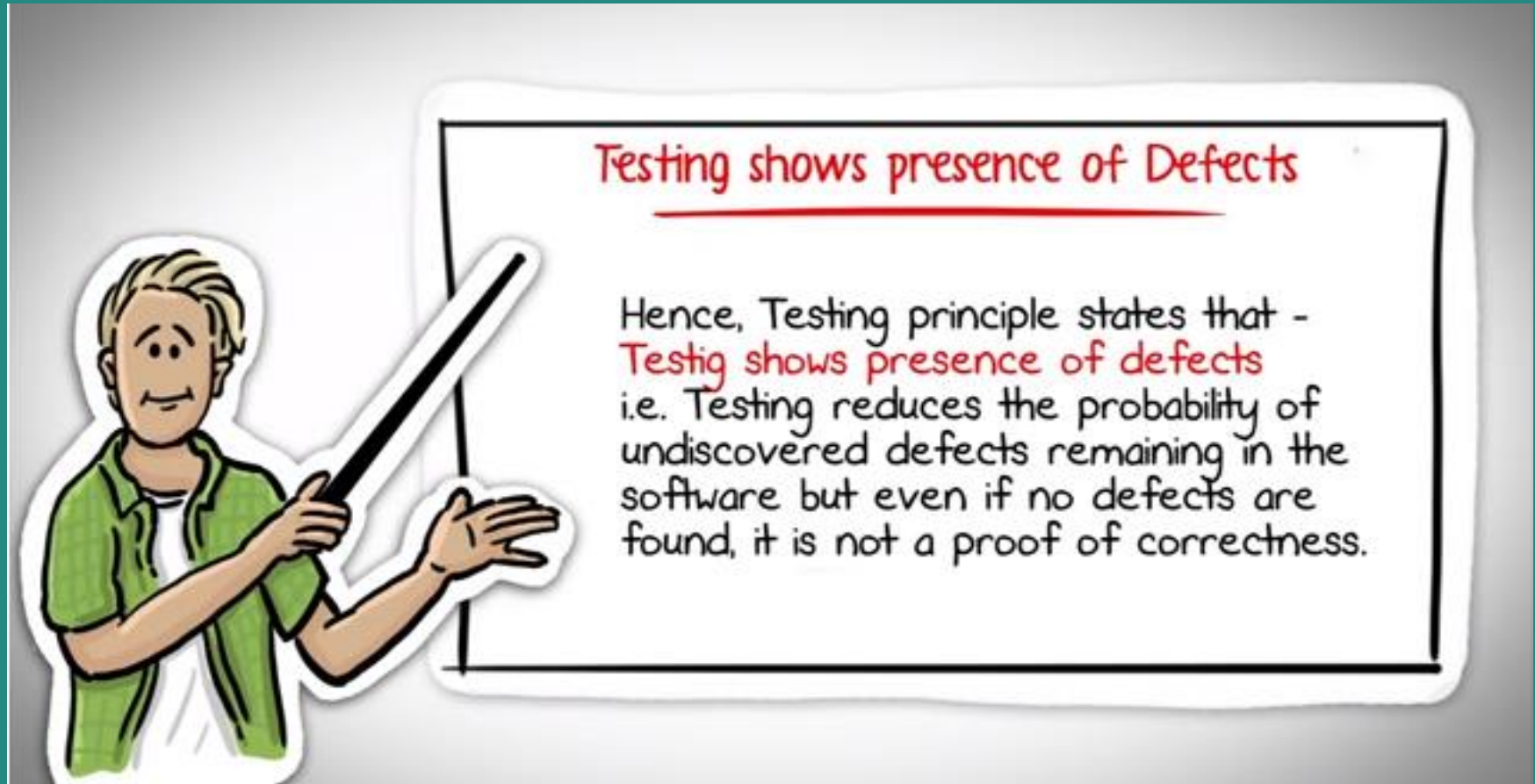
Lỗi xảy ra theo “cụm”

Nghịch lý “thuốc trừ sâu”

Phụ thuộc ngữ cảnh hệ thống

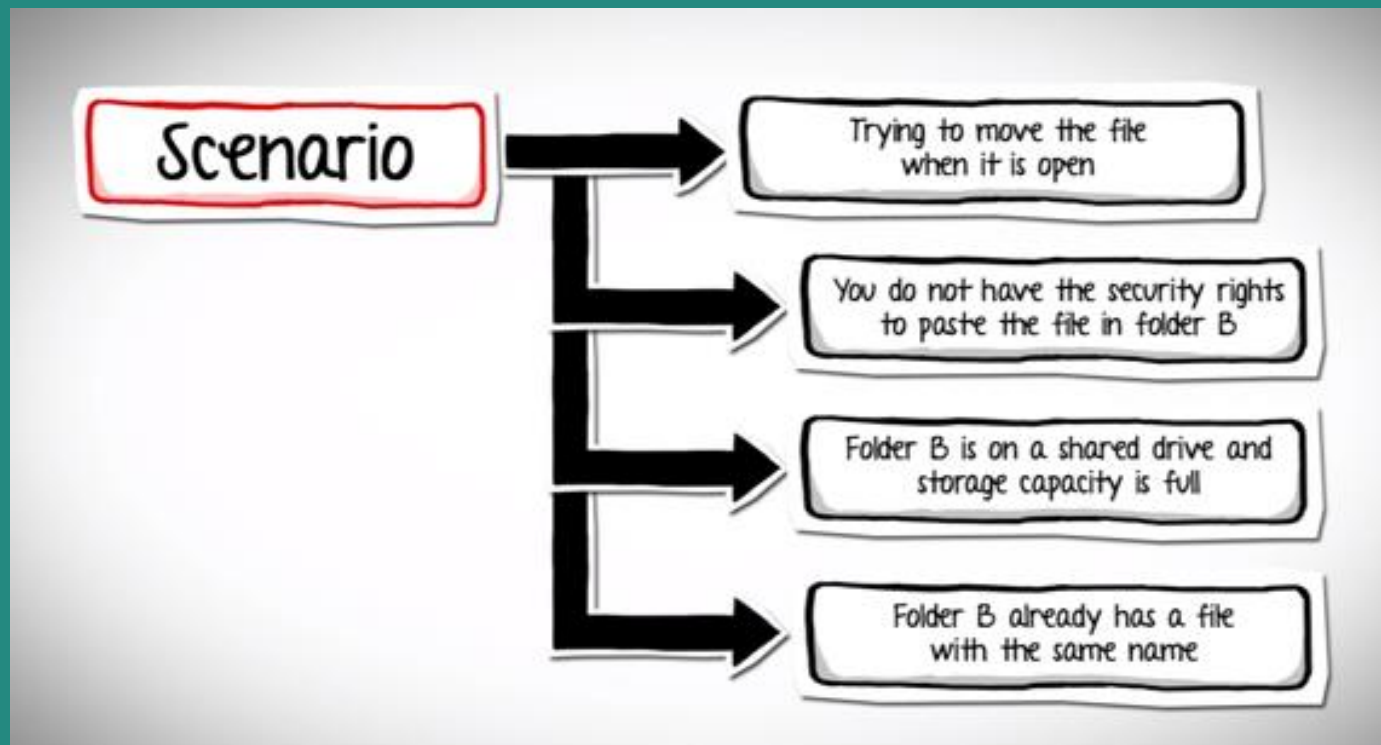
“Ảo tưởng” về việc không lỗi

Testing shows presence of defects



Exhaustive testing is not possible

- A scenario where you are moving a file from folder A to Folder B.



Exhaustive testing is not possible

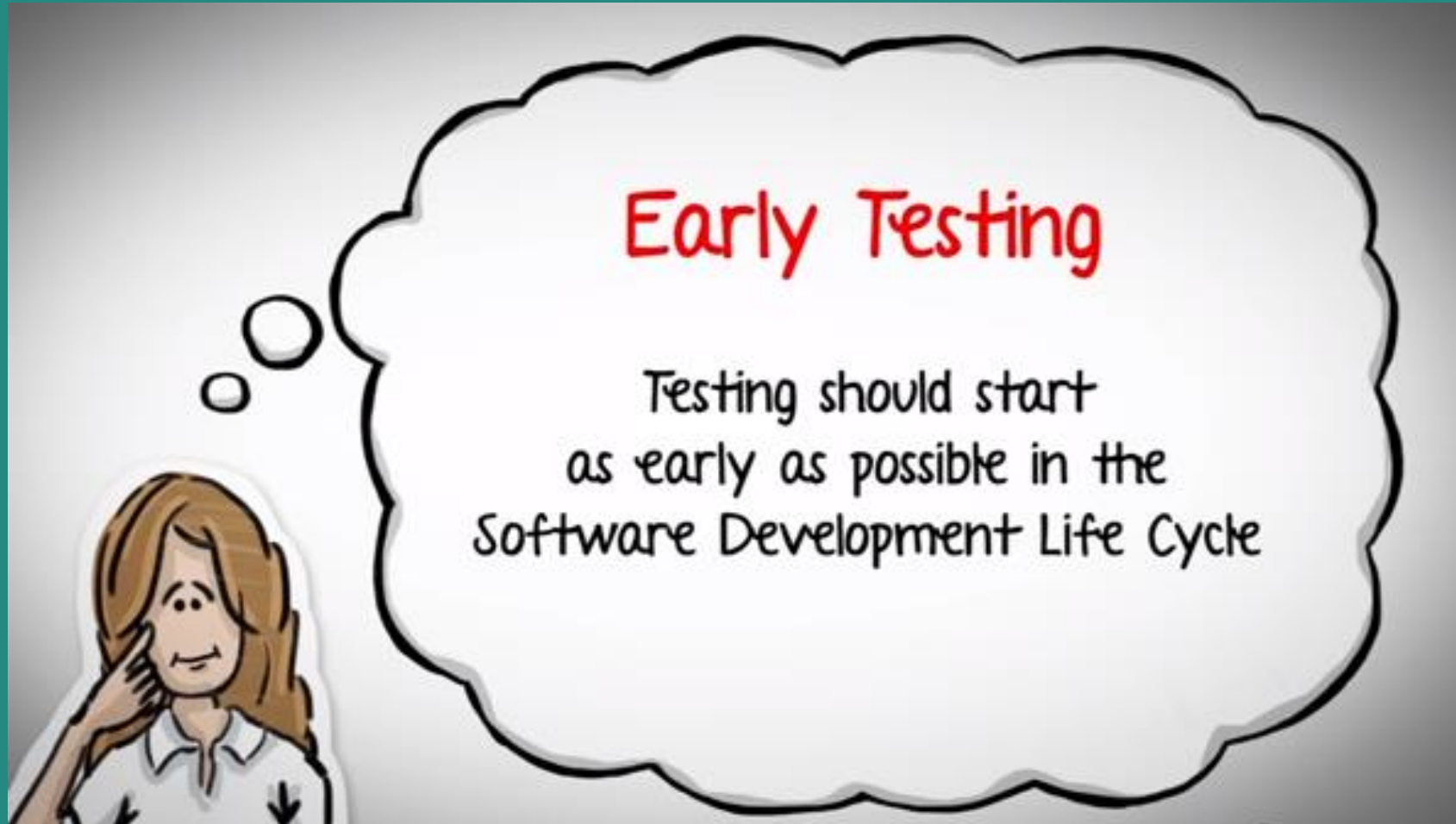
- Instead, we need the optimal amount of testing based on the risk assessment of the application.

✓ Or suppose you have 15 input fields to test each having 5 possible values, the number of combinations to be tested would be $5^{15} = 30\,517\,578\,125!!$



Field 1		Field 2		Field 3	
Field 4		Field 5		Field 6	
Field 7		Field 8		Field 9	
Field 10		Field 11		Field 12	
Field 13		Field 14		Field 15	

Early Testing



Defect clustering

☒ Which operation is most likely to cause your operating system to fail?

A.

opening
Microsoft
Word

B.

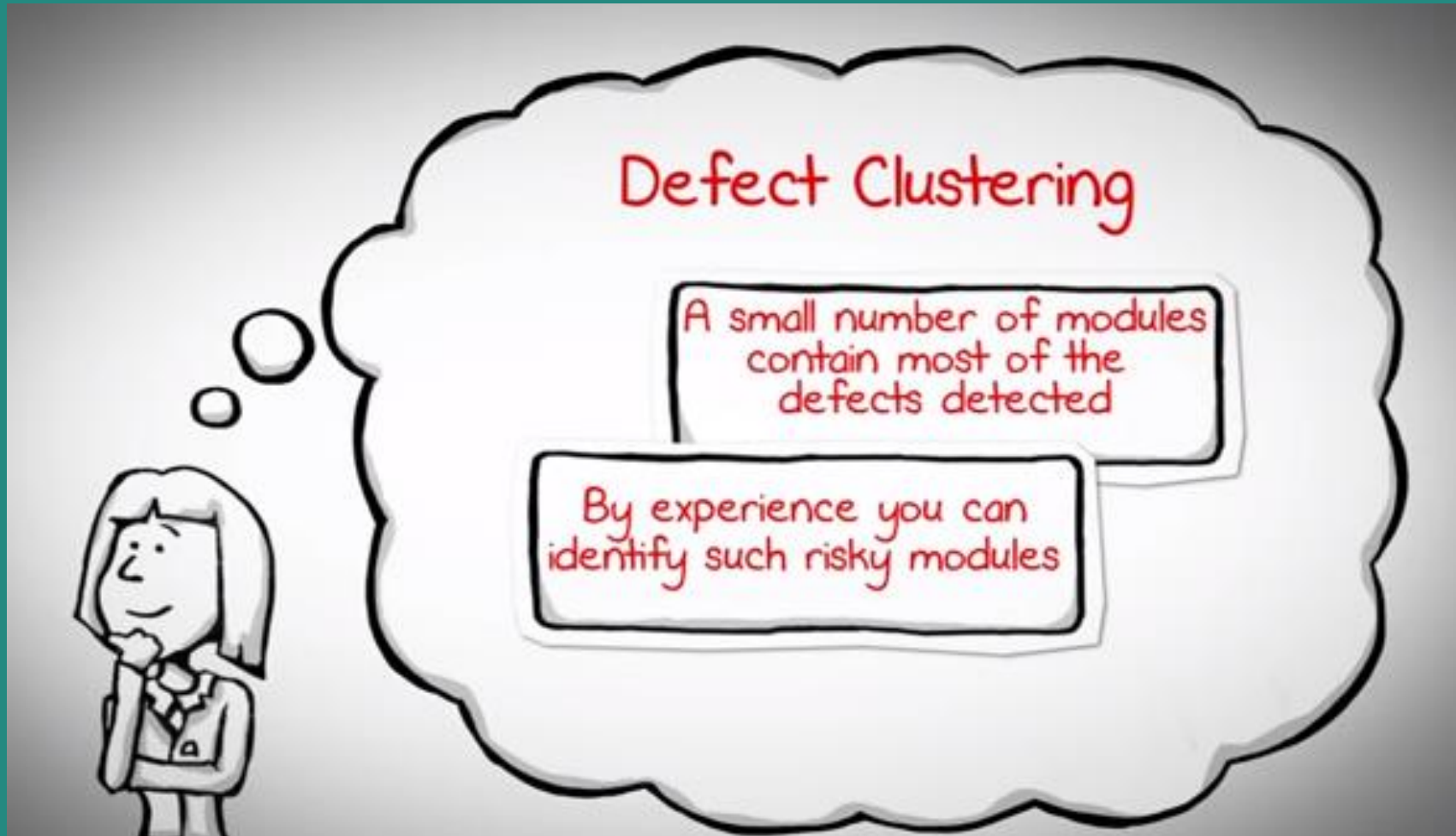
open
Interne
Explorer

C.

opening 10 heavy
graphics
applications all at
the same time

Multi-Tasking

Defect clustering



The testing paradox

Purpose of testing: to find faults

Finding faults destroys confidence

Purpose of testing: destroy confidence

Purpose of testing: build confidence

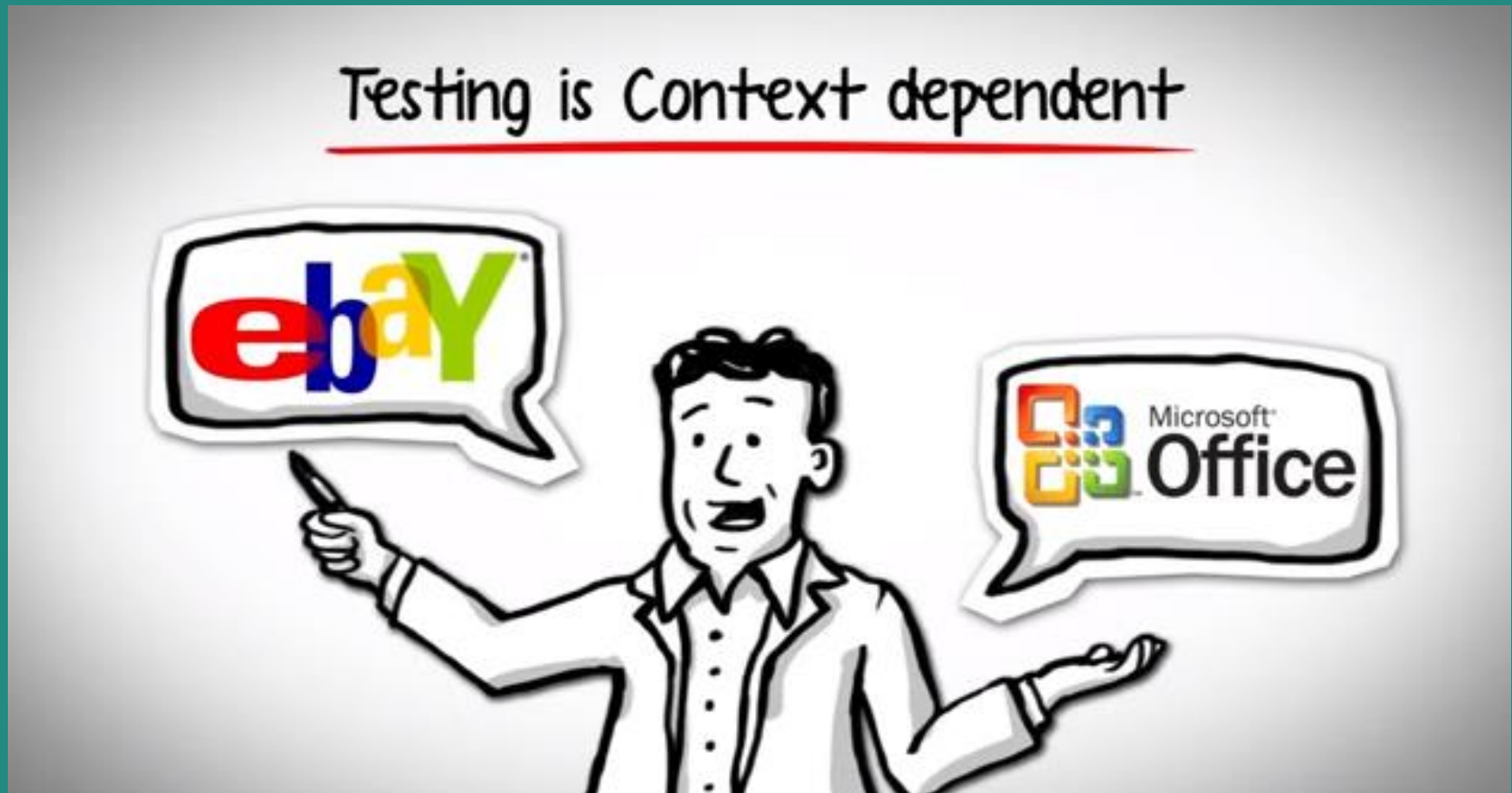
**The best way to build confidence
is to try to destroy it**

Pesticide paradox

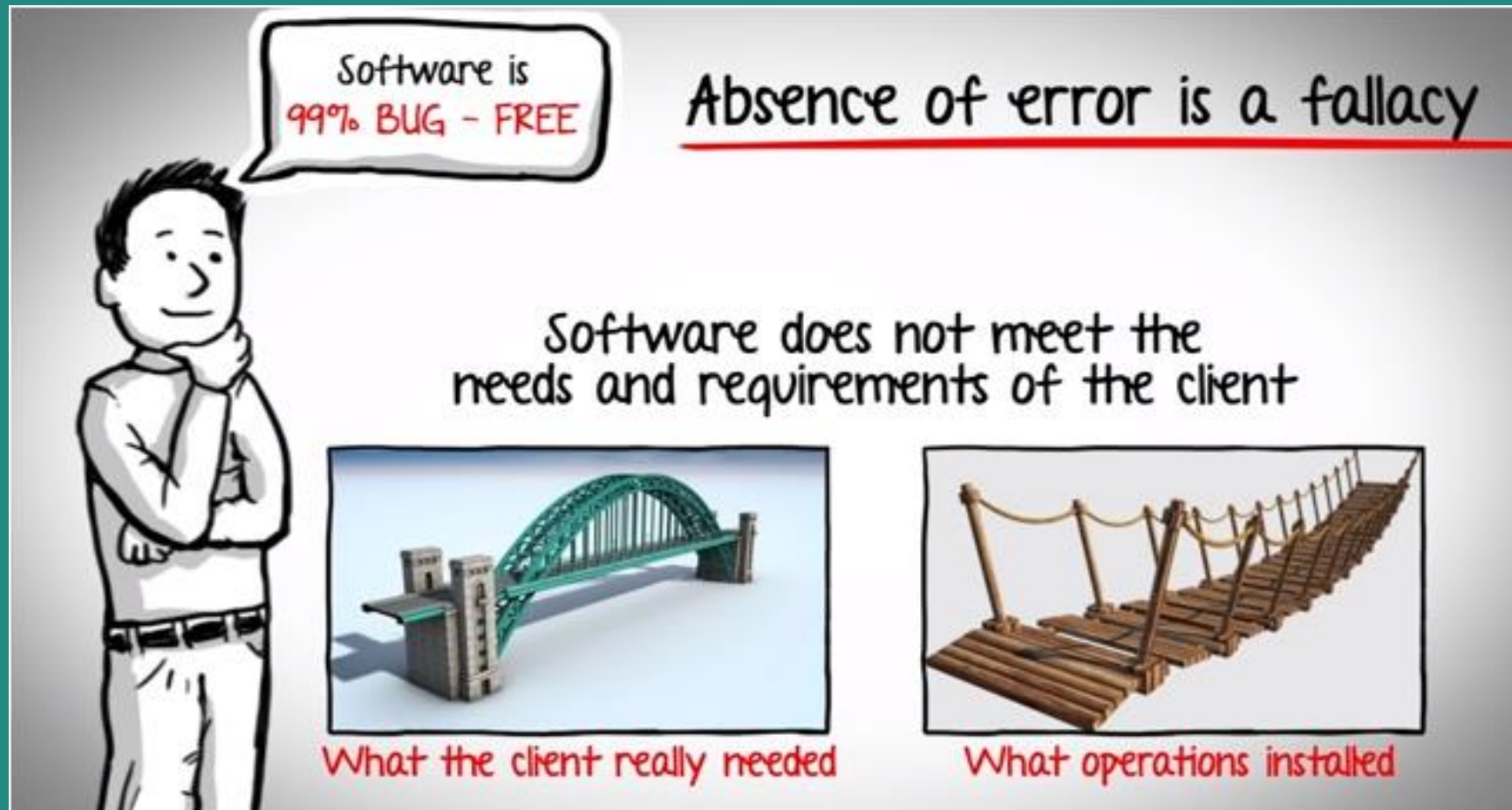
- ☑ If the same tests are repeated over and over again, eventually the same test cases will no longer find new bugs
- ☑ This is the another principle of testing called "Pesticide paradox"
- ☑ To overcome this, the test cases needed to be regularly reviewed & revised, adding new and different test cases to help find more defects.



Testing is context dependent



Absence of errors fallacy



Q&A???

- Hiểu rõ các nguyên tắc trong kiểm thử phần mềm để làm gì?